Reg. No. | | | | | | | | | | | [12ME755]

SEVENTH SEMESTER B.E EXAMINATION, DECEMBER-2015

FOUNDRY TECHNOLOGY

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full Questions.

1.	**a.**	With a sketch, briefly explain what is a hot tear.	**5**
	b.	State the principle of degassing.	**6**
	c.	Give a detailed list of degassing techniques.	**3**
2.	**a.**	List the important designing factors for maximizing casting strength and depict good and bad casting section design.	**7**
	b.	What are ribs? Explain with the aid of a sketch.	**7**
3.	**a.**	Schematically represent segregation.	**3**
	b.	Graphically indicate solidification from a flat mold wall.	**5**
	c.	Indicate temperature distribution in directional solidification.	**6**
4.	**a.**	Indicate correct and incorrect parting line placement schematically.	**6**
	b.	Briefly comment on gating system with the aid of a sketch.	**8**
5.	**a.**	Give the detailed classification of special molding techniques.	**7**
	b.	A cupola of 75 cm diameter has meting ratio of 10:1, calculate **i.** The air required for complete combustion **ii.** Air required to melt 500kg of iron at this ratio **iii.** The coke required to melt 500kg of iron if melting ratio is 8:1 and 10:1 **iv.** The air required ti melt 500kg of iron if melting raito is 8:1 Assume the weight of regular charge of coke as 32 kg.	**7**
6.	**a.**	What is the inspection techniques used ion steel castings?	**5**
	b.	Which are the melting furnaces suitable for melting of steel? Explain direct arc furnace.	**9**
7.	**a.**	Mention the casting properties of aluminum alloys.	**3**
	b.	Which are the melting furnaces suitable for melting of aluminum alloys? Explain pot furnace in detail.	**11**
8.	**a.**	With the flow sheet of sand circulation, schematically indicate a typical mechanized foundry.	**7**
	b.	Give an exhaustive classification of material handling equipment's in a modern foundry.	**7**

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